# A Survey of AI-Driven Protein Design with Diffusion and Language Models

**InteractiveSurvey**


## Abstract

Designing novel proteins with tailored functions represents a transformative challenge in computational biology, recently revolutionized by the convergence of deep learning and expansive structural datasets. This survey critically examines the rapid emergence of AI-driven protein design, specifically focusing on the synergistic integration of diffusion models and protein language models to navigate the vast landscape of protein folds. We provide a systematic taxonomy of current methodologies, detailing how geometric equivariance on Lie groups and flow matching techniques enable the generation of physically valid backbones while respecting intrinsic rotational and torsional constraints. Furthermore, the review analyzes advanced strategies for conditional generation, including force-guided sampling and constraint anchoring, which bridge statistical learning with biophysical reality to ensure functional viability and experimental testability. By synthesizing recent algorithmic advancements in joint sequence-structure design, this work identifies key open challenges and future directions, serving as an essential resource for developing the next generation of generative models for de novo protein engineering.


## 1 Introduction

The quest to design novel proteins with tailored functions represents one of the most profound challenges in modern computational biology, holding transformative potential for therapeutics, enzymatic catalysis, and materials science. Traditionally, protein engineering relied heavily on directed evolution or rational design, methods often constrained by the limited exploration of sequence space and an incomplete understanding of the complex mapping between amino acid sequences and three-dimensional structures. The recent explosion of structural data, coupled with breakthroughs in deep learning, has ushered in a new era where generative artificial intelligence can navigate the vast landscape of protein folds [1]. These data-driven approaches promise to transcend natural evolutionary constraints, enabling the creation of entirely synthetic biomolecules with unprecedented stability and functionality, thereby shifting the paradigm from analyzing existing proteins to inventing new ones.

This survey paper focuses specifically on the rapid emergence of AI-driven protein design powered by diffusion models and protein language models. While earlier generative approaches utilized variational autoencoders or generative adversarial networks, diffusion models have recently established themselves as the state-of-the-art framework for modeling the complex, high-dimensional distributions of protein structures and sequences [2]. By simulating a reverse noise-to-structure process, these models offer superior sample quality and mode coverage compared to their predecessors. Furthermore, the integration of protein language models provides rich evolutionary priors, enhancing the biological plausibility of generated sequences [3]. This work critically examines the synergy between geometric deep learning, which ensures physical validity, and probabilistic generative modeling, which drives structural innovation.

The core of this review begins by dissecting the mathematical foundations of diffusion processes adapted for protein geometry, emphasizing the critical role of geometric equivariance. We explore how SE(3) invariant frameworks model backbones as sequences of local reference frames to ensure consistency under rigid body transformations, eliminating the need for data augmentation. The discussion extends to specialized diffusion mechanisms on Lie groups, such as SO(3) for rotations and toroidal manifolds for torsion angles, which respect the intrinsic curvature of rotational variables. Additionally, we analyze flow

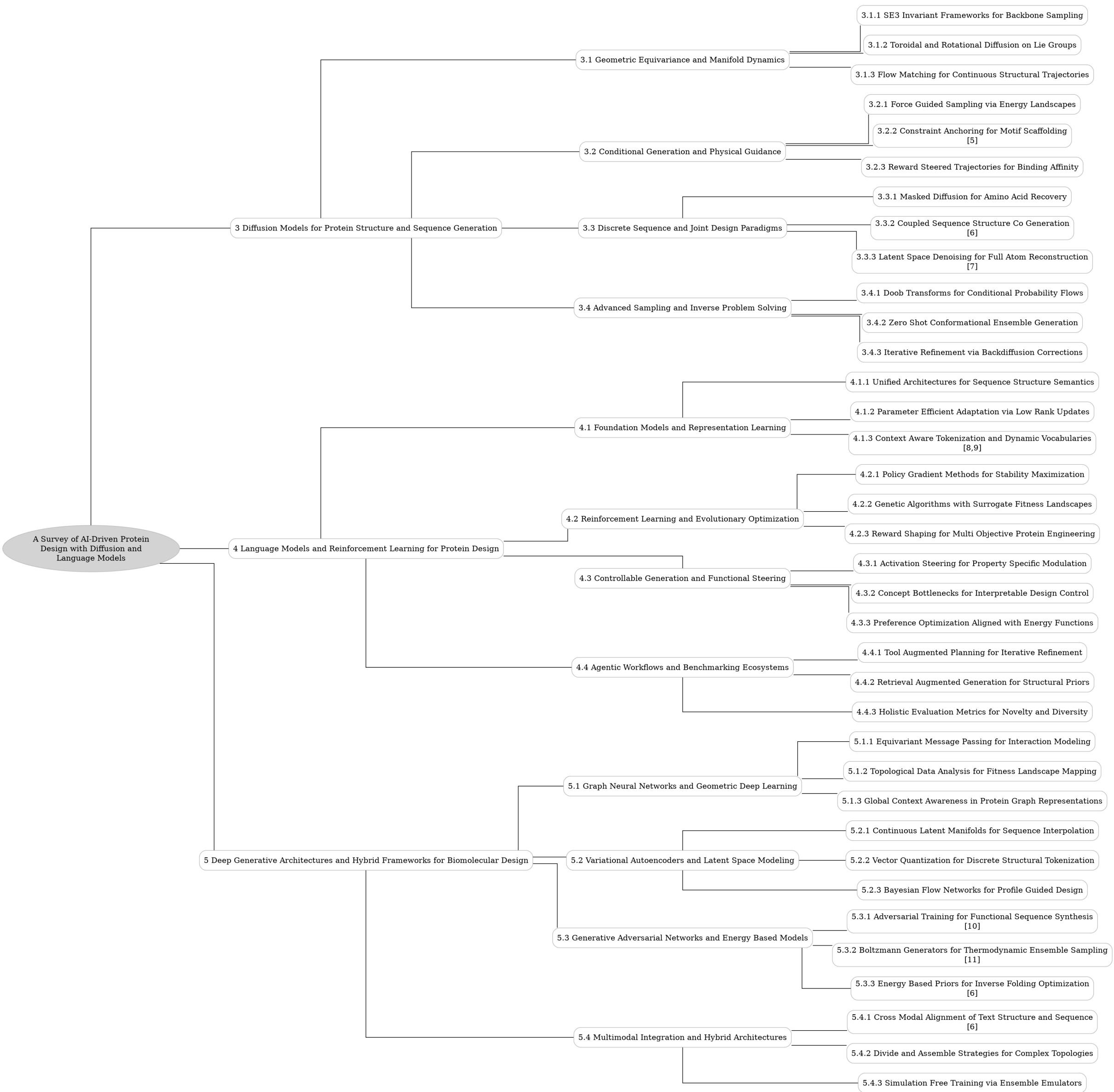


Figure 1: The outline of our survey: A Survey of AI-Driven Protein Design with Diffusion and Language Models

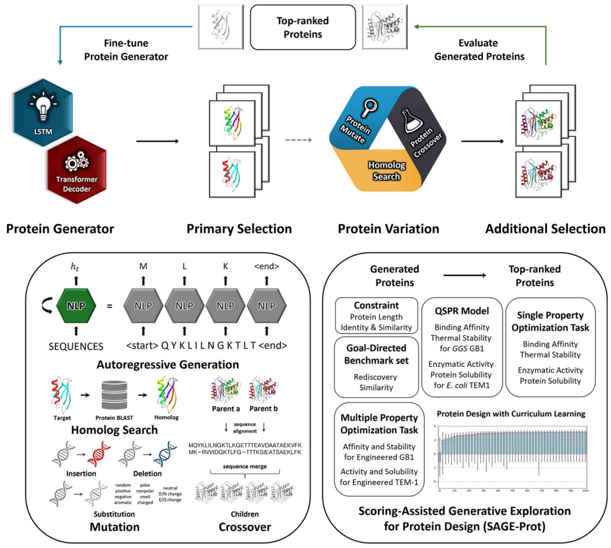


Figure 2: Chart from 'Scoring-Assisted Generative Exploration for Proteins (SAGE-Prot): A Framework for Multi-Objective Protein Optimization via Iterative Sequence Generation and Evaluation' (arXiv:2505.01277)

$$p(x) = \int p(x|z)p(z)dz$$

Figure 3: Chart from 'Protein sequence design with deep generative models' (arXiv:2104.04457)

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

Figure 4: Chart from 'BERT AND LLMS-BASED AVGFP BRIGHTNESS PREDICTIONAND MUTATION DESIGN' (arXiv:2407.20534)

matching techniques that provide simulation-free, deterministic trajectories for continuous structural generation, offering a computationally efficient alternative to stochastic differential equations while maintaining high fidelity in backbone synthesis.

Subsequently, the survey investigates advanced strategies for conditional generation and physical guidance, which are essential for practical design tasks. We detail force-guided sampling methods that integrate auxiliary energy functions to steer generative trajectories toward low-energy, physically plausible conformations, effectively bridging statistical learning with biophysical reality. The review also covers constraint anchoring techniques for motif scaffolding, where functional substructures are rigidly fixed to solve complex inpainting problems, and reward-steered trajectories that optimize binding affinity during the denoising process. These sections highlight how modern architectures incorporate hard geometric constraints and soft physical potentials to produce designs that are not only novel but also functionally viable and experimentally testable.

Finally, the paper addresses the intricate challenge of joint sequence-structure design, moving beyond backbone-only generation to encompass discrete amino acid recovery. We examine masked diffusion paradigms that treat sequence assignment as a discrete denoising task conditioned on fixed structures, leveraging geometric constraints to recover optimal side-chain identities. This comprehensive overview synthesizes recent algorithmic advancements, comparing their theoretical underpinnings and empirical performance across diverse design benchmarks. By connecting the dots between continuous structural generation and discrete sequence optimization, this survey provides a unified perspective on the current capabilities and limitations of generative models in protein science [1].

The primary contribution of this work is a systematic and critical synthesis of the rapidly evolving landscape of diffusion-based protein design. Unlike previous reviews that focus broadly on deep learning in biology, this paper offers a deep technical dive into the specific architectural innovations required to handle the unique geometric and discrete nature of proteins. We provide a structured taxonomy of existing methods, categorizing them by their handling of geometric equivariance, conditioning mechanisms, and sequence-structure coupling. Furthermore, we identify key open challenges and future research directions, serving as an essential resource for researchers aiming to develop the next generation of generative models for de novo protein engineering [4].

## 2 Diffusion Models for Protein Structure and Sequence Generation

### 2.1 Geometric Equivariance and Manifold Dynamics

#### 2.1.1 SE3 Invariant Frameworks for Backbone Sampling

SE(3) invariant frameworks have emerged as a foundational paradigm for protein backbone generation, addressing the critical need for geometric consistency under rigid body transformations. By modeling backbones as sequences of local reference frames rather than absolute Cartesian coordinates, these approaches ensure that the learned probability distributions remain invariant to global translation and rotation. This mathematical formulation leverages the properties of the SE(3) Lie group to define diffusion processes or flow matching trajectories directly on the manifold of valid protein structures, thereby eliminating the need for data augmentation and improving sample efficiency.

Recent advancements implement these principles through equivariant denoising diffusion probabilistic models and stochastic flow matching techniques. In these architectures, the generative process iteratively refines noisy frame configurations into coherent tertiary structures by predicting updates to rotation matrices and translation vectors simultaneously. The integration of SE(3)-equivariant neural networks allows the model to capture complex spatial dependencies between residues while strictly preserving physical constraints such as bond lengths and angles. Furthermore, specific implementations incorporate rigid-body preprocessing steps to maintain chirality, mitigating the generation of non-physical left-handed helices that plague coordinate-based methods.

The efficacy of SE(3) invariant sampling is particularly evident in conditional generation tasks, such as motif scaffolding and joint sequence-structure design. These frameworks facilitate the incorporation of structural priors, enabling the model to generate diverse backbones that satisfy specific geometric constraints without compromising foldability. Empirical evaluations demonstrate that backbones produced via SE(3) diffusion exhibit higher designability scores when paired with sequence design tools, as the generated structures possess energy landscapes conducive to stable folding. Consequently, these invariant frameworks represent a significant shift from heuristic sampling toward principled, geometry-aware generative modeling in computational protein design.

#### 2.1.2 Toroidal and Rotational Diffusion on Lie Groups

Standard Euclidean diffusion frameworks are insufficient for modeling rotational variables and torsion angles, as these entities reside on compact Riemannian manifolds rather than flat vector spaces. To address this geometric mismatch, recent methodologies extend diffusion processes to Lie groups, specifically utilizing the special orthogonal group SO(3) for global orientations and the torus for periodic dihedral angles. This adaptation requires replacing Gaussian noise injection with operations defined by the heat kernel on compact Lie groups, ensuring that the forward noising process respects the intrinsic curvature and topological constraints of the underlying manifold.

For rotational diffusion on SO(3), the forward process is often modeled as Brownian motion on the group, where the transition kernel corresponds to the heat kernel derived from the Laplace-Beltrami operator. While theoretical formulations rely on repeated applications of this kernel to diffuse rotations toward a uniform prior, practical implementations frequently employ spherical linear interpolation (SLERP) between the initial state and samples drawn from the uniform distribution on SU(2) or SO(3). This approach effectively navigates the non-Euclidean geometry by interpolating along geodesics, thereby maintaining valid rotation matrices throughout the diffusion trajectory without violating group properties.

Similarly, torsion angles exhibiting periodicity are treated as points on a toroidal manifold, necessitating the use of wrapped normal distributions to handle boundary conditions within the interval $[-\pi, \pi)$. The reverse denoising process in these settings utilizes equivariant score matching, where the predicted score function is constrained to be invariant under the group action, ensuring that the generated structures preserve physical symmetries. By integrating these geometric priors directly into the diffusion architecture, models can generate physically realizable protein conformations and molecular structures that strictly adhere to the rigid body constraints inherent in 3D spatial reasoning.

#### 2.1.3 Flow Matching for Continuous Structural Trajectories

Flow matching has emerged as a simulation-free alternative to traditional diffusion for modeling continuous structural trajectories in protein design. By directly learning the vector field that guides the optimal probability path from a prior noise distribution to the target data distribution, this framework eliminates the need for explicit knowledge of the marginal vector field required by Continuous Normalizing Flows. Unlike standard diffusion processes that rely on stochastic differential equations, flow matching formulates generation as a deterministic Ordinary Differential Equation (ODE), enabling high-fidelity structure synthesis with significantly fewer integration steps. This efficiency is particularly critical when navigating the complex, high-dimensional manifolds of protein backbones, where rapid convergence to stable conformations is essential for practical application.

Recent advancements extend these principles to the special Euclidean group SE(3), addressing the geometric constraints of rigid body motions inherent in residue frames. Models such as FrameFlow leverage flow matching over SE(3) to jointly model rotations and translations, ensuring equivariance without the computational overhead of complex score networks. By operating on Frenet-Serret frames or separate rotation-translation components, these methods capture the global dependencies of residue-wise interactions more effectively than autoregressive approaches, which often struggle with ambiguous volumes and require extensive backtracking. The deterministic nature of the flow allows for smooth interpolation between structural states, facilitating the generation of continuous trajectories that respect physical plausibility and geometric consistency.

Furthermore, flow matching enables sophisticated conditional generation tasks, such as motif scaffolding and partial structure inpainting, by conditioning the vector field on specific structural constraints. Techniques like rectified flow further optimize the trajectory, straightening the probability path to accelerate sampling while maintaining diversity. When applied to continuous structural coordinates, this approach mitigates the failure modes associated with discrete step-wise denoising, allowing for the seamless integration of sequence and structure information. Consequently, flow matching provides a robust mathematical foundation for generating novel protein architectures, offering a scalable pathway to explore the vast landscape of native sequences and their corresponding three-dimensional folds.

### 2.2 Conditional Generation and Physical Guidance

#### 2.2.1 Force Guided Sampling via Energy Landscapes

Force-guided sampling leverages auxiliary energy functions to navigate the complex conformational space of biomolecules, offering fine-grained control beyond standard stochastic diffusion. By incorporating gradients from physical potentials, such as those penalizing deviations from specific distance constraints or torsional angles, the generative process is actively steered toward low-energy regions. This approach transforms the reverse diffusion trajectory from a purely data-driven denoising path into a physics-informed optimization, ensuring that intermediate states remain physically plausible while satisfying user-defined structural criteria without requiring model retraining.

The integration of these energy landscapes often manifests through modified sampling algorithms that blend learned score functions with explicit force fields. Techniques such as Langevin dynamics or geodesic random walks are employed to discretize the update steps, where the drift term is augmented by the negative gradient of the auxiliary energy function. This hybrid formulation allows for the correction of geometric inconsistencies, such as steric clashes or unrealistic bond lengths, directly during the generation phase. Furthermore, methods utilizing particle filtering or sequential Monte Carlo strategies enable the exploration of diverse conformers by maintaining an ensemble of trajectories that are continuously reweighted based on their adherence to the target energy landscape.

Recent advancements extend this paradigm to handle periodic representations, such as internal angles, and complex topological constraints like TIM-barrel folds. By interpolating between uniform random rotations and target configurations via spherical linear interpolation, models can maintain well-defined diffusion processes even in highly constrained manifolds. The ability to condition generation on experimental data, such as distance restraints derived from CryoEM or NMR, further enhances the fidelity of the sampled ensembles. Ultimately, force-guided sampling bridges the gap between statistical learning and biophysical reality, facilitating the design of proteins with precise mechanical properties and functional capabilities.

### 2.2.2 Constraint Anchoring for Motif Scaffolding

Constraint anchoring serves as the fundamental mechanism for solving the motif-scaffolding problem by rigidly fixing functional substructures within a generative diffusion or flow matching framework. In this paradigm, specific residue frames corresponding to a target motif are held invariant throughout the denoising trajectory, effectively acting as hard geometric boundary conditions. The generative model, typically an $SE(3)$-equivariant network, is tasked with synthesizing the surrounding scaffold backbone that seamlessly connects these anchored regions while maintaining physical plausibility and avoiding steric clashes. This approach transforms the unconditional generation of novel folds into a conditional inpainting task, where the probability distribution is restricted to structures compatible with the predefined functional geometry.

The technical implementation of anchoring often involves modifying the noise prediction or velocity field estimation to respect the fixed coordinates of the motif residues. During each iterative step of the reverse process, the generated coordinates for the anchored frames are explicitly overwritten with their ground-truth values, ensuring zero deviation from the target conformation. Advanced methodologies extend this basic constraint by incorporating gradient-based guidance or energy terms that penalize discontinuities at the motif-scaffold junctions. This ensures not only that the motif remains static but also that the transition regions exhibit smooth backbone torsion angles and appropriate bond lengths, thereby preserving the local chemical environment required for the motif's biological activity.

Recent advancements have expanded constraint anchoring beyond single static motifs to support multi-motif scaffolding without requiring explicit specification of inter-motif spatial arrangements [5]. By anchoring multiple distinct functional sites simultaneously, the model learns to infer feasible relative orientations and distances that satisfy global foldability constraints. This capability allows for the design of complex proteins featuring co-occurring binding sites or catalytic residues derived from different parent structures. The success of these methods relies heavily on the model's capacity to navigate the high-dimensional structural landscape to find low-energy solutions that satisfy all geometric constraints while maximizing structural diversity and novelty in the generated scaffolds.

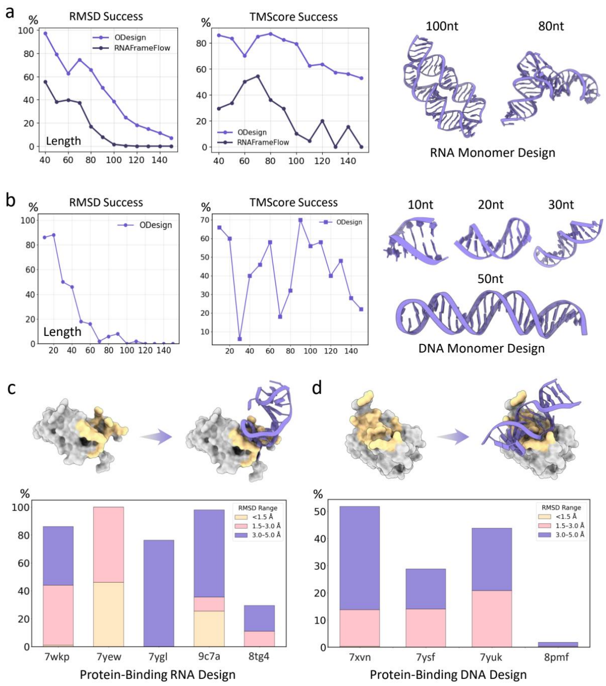


Figure 5: Chart from 'ODesign: A World Model for Biomolecular Interaction Design' (arXiv:2510.22304)

### 2.2.3 Reward Steered Trajectories for Binding Affinity

Recent advancements in generative modeling have shifted towards reward-steered trajectories to explicitly optimize binding affinity during the diffusion process. By integrating differentiable scoring functions or classifier guidance directly into the denoising steps, models can bias the generation of protein-ligand complexes toward thermodynamically favorable states. This approach allows for the continuous refinement of interface geometry, ensuring that the resulting conformations maximize shape complementarity and electrostatic compatibility without relying solely on post-hoc filtering.

Techniques such as PAE-guided and long-range contact-guided trajectories demonstrate how specific structural constraints can be encoded early in the generation timeline to enforce critical interaction motifs. For instance, guiding the diffusion path using predicted aligned error metrics encourages the formation of short-range contacts initially, while alternative guidance signals promote distal interactions essential for stable complex assembly. Furthermore, incorporating potentials derived from experimental observables or physics-based energy functions enables the model to navigate the conformational landscape more effectively, addressing the challenges of induced fit and flexible binding sites.

The efficacy of these steered methods is evidenced by the successful design of binders exhibiting picomolar affinities against diverse targets, including helical peptides and viral surface proteins. By jointly optimizing backbone coordinates and interface residues through reinforcement learning or adaptive noise schedules, these frameworks overcome the limitations of static docking assumptions. Consequently, reward-steered diffusion represents a pivotal evolution in de novo design, transforming the generation process from a purely structural exercise into a function-driven optimization capable of producing high-affinity therapeutic candidates with unprecedented precision.

## 2.3 Discrete Sequence and Joint Design Paradigms

### 2.3.1 Masked Diffusion for Amino Acid Recovery

Masked diffusion for amino acid recovery operates by treating sequence design as a discrete denoising task conditioned on fixed structural backbones. Unlike continuous diffusion models that perturb Cartesian coordinates, this approach initializes the sequence with a fully masked state or Gaussian noise applied to one-hot encoded residues, where true amino acids are marked as 1 and others as -1. The generative process iteratively refines these latent representations, leveraging the geometric constraints of the input structure to guide the probability distribution toward biologically viable sequences. This methodology effectively decouples backbone generation from sequence assignment, allowing the model to focus exclusively on recovering optimal side-chain identities that satisfy local steric and global energetic requirements.

The architectural backbone of these models frequently integrates SE(3)-equivariant modules, such as Invariant Point Attention (IPA) blocks, to ensure that sequence predictions remain consistent regardless of the protein's spatial orientation. By conditioning the denoising network on noised sequence states and intermediate structural features, the model learns to predict corrected residue types that align with amino-acid specific distributions inherent in natural proteins. This framework avoids reliance on explicit biophysical potentials during training, instead internalizing statistical priors directly from data. The reverse diffusion process gradually transitions the system from a high-entropy masked distribution to a low-entropy, deterministic sequence, effectively solving the inverse folding problem through probabilistic refinement rather than autoregressive token prediction.

Recent advancements have extended this paradigm to include multi-level conditional diffusion, simultaneously optimizing backbone frames and discrete amino acid identities to enhance design accuracy and novelty. Models employing this strategy demonstrate superior capability in capturing long-range dependencies and complex residue interactions compared to traditional autoregressive methods. By utilizing discrete diffusion extensions, these systems provide a structured iterative refinement process that yields raw outputs requiring minimal post-processing. The efficacy of masked diffusion is particularly evident in tasks such as motif scaffolding and rotamer repacking, where the model successfully recovers functional sequences that maintain the stability of the prescribed structural topology.

### 2.3.2 Coupled Sequence Structure Co Generation

Coupled sequence-structure co-generation represents a paradigm shift from traditional sequential pipelines, addressing the inherent limitations of decoupled backbone sampling and fixed-backbone sequence design. Unlike earlier methods that ignore critical ligand-side chain and backbone-side chain interactions during initial structure formation, co-generative frameworks simultaneously model amino acid identities and their corresponding three-dimensional coordinates. This joint probability modeling ensures that the generated sequences are intrinsically compatible with the evolving structural geometry, thereby mitigating the risk of producing non-foldable scaffolds that often plague approaches relying on post-hoc inverse folding [6].

$$\rho_i = \sqrt{\frac{1}{C}\mathrm{Tr}([H^{-1}]_{ii})},$$

Figure 6: Chart from 'Learning to Engineer Protein Flexibility' (arXiv:2412.18275)

Technically, these models frequently leverage SE(3)-equivariant diffusion processes or autoregressive transformers to operate within a unified latent space where sequence and structure tokens are interdependent. By incorporating rigid-body rotation-invariant preprocessing and torsion angle parameters, the architecture maintains geometric consistency while sampling discrete amino acid types alongside continuous atomic positions. This formulation allows the model to capture complex epistatic relationships and co-evolutionary signals directly, enabling the synthesis of macromolecular complexes and nucleic acid-protein interfaces that require precise inter-residue coordination unachievable through isolated module optimization.

The efficacy of co-generation is primarily validated through self-consistency metrics, such as self-consistency RMSD and TM-score, which measure the structural agreement between the generated conformation and the structure predicted from the generated sequence. Empirical evaluations demonstrate that jointly optimized pairs exhibit superior designability and higher sequence recovery rates compared to those derived from hallucination or reinforcement learning baselines. Furthermore, this approach facilitates controllable generation modes, including motif scaffolding and loop engineering, by conditioning the coupled diffusion process on specific structural constraints while preserving the physical plausibility of the entire protein fold.

#### 2.3.3 Latent Space Denoising for Full Atom Reconstruction

Latent space denoising addresses the computational complexity of generating high-resolution full-atom protein structures by operating within a compressed, continuous representation rather than directly manipulating atomic coordinates. This paradigm typically employs a variational autoencoder to map discrete amino acid sequences and geometric backbone frames into a smoothed latent manifold, effectively decoupling sequence semantics from structural geometry [7]. By diffusing noise within this lower-dimensional space, models can capture long-range dependencies and global topological constraints more efficiently than atom-level approaches, significantly reducing the dimensionality of the generative process while preserving essential physicochemical properties.

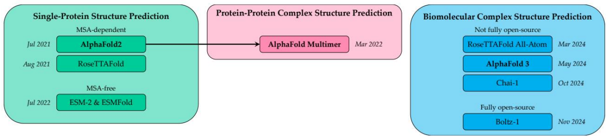


Figure 7: Chart from 'A Model-Centric Review of Deep Learning for Protein Design' (arXiv:2502.19173)

To ensure physical validity during the reverse denoising trajectory, these frameworks integrate SE(3)-equivariant architectures that respect the rotational and translational symmetries inherent to molecular systems. The denoiser predicts updates to local coordinate frames defined by C-alpha atoms, utilizing invariant point attention mechanisms to maintain geometric consistency across iterative refinement steps. This equivariance allows the model to learn a canonical distribution of structural motifs independent of global orientation, enabling the reconstruction of precise bond lengths, angles, and torsion parameters required for all-atom fidelity without violating steric constraints or chemical valency rules.

The final reconstruction phase involves a deterministic decoding step that maps the refined latent vectors back to explicit atomic coordinates, often incorporating side-chain packing algorithms to resolve rotamer conformations. Advanced implementations utilize hierarchical conditional flows, where information propagates from coarse backbone representations to fine-grained atomic details, ensuring that local atomic environments remain consistent with the global fold. This multiscale approach facilitates the generation of novel, stable protein structures with high structural accuracy, effectively bridging the gap between abstract latent representations and experimentally viable full-atom models suitable for downstream functional design.

### 2.4 Advanced Sampling and Inverse Problem Solving

#### 2.4.1 Doob Transforms for Conditional Probability Flows

Doob transforms provide a rigorous mathematical framework for constructing conditional probability flows by modifying the drift term of an underlying stochastic differential equation. In the context of generative modeling, this technique allows for the exact conditioning of a diffusion process on terminal states or auxiliary observations without requiring retraining of the base model. By applying an h-transform to the unconditional transition kernel, the resulting process is guided towards the desired constraint manifold, effectively turning an intractable conditional sampling problem into a tractable simulation of a modified stochastic dynamics system.

The implementation of Doob transforms involves estimating the gradient of the log-conditional probability, often referred to as the conditional score, which acts as an additional force in the reverse-time dynamics. This approach decomposes the target conditional distribution into the product of the learned unconditional density and a likelihood term derived from the conditioning signal. Consequently, the reverse process evolves under a new drift coefficient that combines the original denoising score with the gradient of the log-likelihood of the condition, ensuring that sampled trajectories remain consistent with the specified constraints throughout the generation horizon.

This methodology is particularly advantageous for complex structured data, such as protein backbones defined on SE(3) manifolds, where direct conditioning via standard classifier guidance may introduce approximation errors. Unlike heuristic guidance scales that can distort the data distribution, Doob transforms theoretically preserve the correct conditional marginal distributions at every timestep by strictly adhering to the Girsanov theorem. This ensures that the generated samples not only satisfy the geometric or functional constraints but also maintain the high-fidelity statistical properties of the native data distribution, offering a robust solution for precise molecular design tasks.

#### 2.4.2 Zero Shot Conformational Ensemble Generation

Zero-shot conformational ensemble generation addresses the critical limitation of static structure prediction by modeling the intrinsic dynamism of biomolecules without task-specific fine-tuning. Unlike traditional stochastic search algorithms reliant on handcrafted energy functions, modern approaches leverage deep generative models to directly sample from the underlying Boltzmann distribution. These methods aim to capture multiple metastable states and intrinsically disordered regions, moving beyond the single-structure frontier established by earlier deterministic predictors. By learning the complex energy landscape from vast structural databases, these models can generate diverse, physically plausible conformations that reflect thermal fluctuations and functional heterogeneity inherent in native proteins.

A dominant paradigm in this domain involves equivariant diffusion models operating within SE(3) or torsional angle spaces. Geometric diffusion frameworks, such as GeoDiff and Torsional Diffusion, iteratively denoise random noise into valid 3D coordinates or internal angles while strictly preserving rotational and translational symmetries. This formulation allows for the direct generation of conformational ensembles by varying the initial noise seeds or manipulating the diffusion trajectory, effectively approximating molecular dynamics simulations at a fraction of the computational cost. Furthermore, recent advancements extend these capabilities to intrinsically disordered proteins by adjusting inference-time diffusion steps, enabling the unified modeling of both folded domains and flexible loops within a single architecture.

Despite significant progress, challenges remain in ensuring the physical fidelity and diversity of generated ensembles. A key benchmark involves distinguishing between proteins that genuinely possess multiple conformational states and those where the model erroneously predicts spurious heterogeneity due to training data biases. Current research focuses on refining score networks to better approximate the true Stein score of the conformational distribution, thereby improving the sampling efficiency of rare but biologically relevant states. Future directions emphasize integrating physics-informed constraints and developing robust evaluation metrics to validate that generated ensembles accurately represent the thermodynamic properties and functional mechanisms of dynamic biomolecular systems.

#### 2.4.3 Iterative Refinement via Backdiffusion Corrections

Iterative refinement via backdiffusion corrections addresses the challenge of generating physically plausible protein structures by progressively correcting errors accumulated during the initial denoising phases. Unlike standard diffusion models that rely on a single forward pass, this approach employs a feedback loop where intermediate structural predictions are re-noised and subsequently re-denoised to align more closely with physical constraints and adjacency priors. This mechanism is particularly critical for tasks requiring high fidelity, such as full-atom loop generation and motif scaffolding, where minor deviations in backbone coordinates can lead to steric clashes or unstable conformations.

The technical implementation often leverages SE(3)-equivariant architectures to ensure that rotational and translational symmetries are preserved throughout the correction cycles. By utilizing pretrained weights from structure prediction networks, these methods compute gradient differences between true and predicted denoised frames to guide the refinement trajectory. This process effectively mitigates exposure bias, as the model continuously updates its internal representation of the sequence and structure pair at each time step. Consequently, the system can dynamically adjust side-chain placements and backbone torsions, ensuring that the final output maintains rigorous geometric consistency without requiring extensive retraining on specific structural motifs.

Empirical evaluations indicate that such iterative strategies significantly enhance designability and structural self-consistency across varying protein lengths compared to non-iterative baselines. While increasing the number of refinement iterations improves accuracy, it introduces computational overhead and challenges in managing intermediate state stability. To balance efficiency and performance, recent advancements integrate accelerated sampling techniques, such as rectified flow matching, which straighten the probability path to achieve high-quality results in fewer steps. This optimization allows the model to converge rapidly to native-like solutions, making iterative backdiffusion a viable strategy for large-scale de novo protein design applications.

## 3 Language Models and Reinforcement Learning for Protein Design

### 3.1 Foundation Models and Representation Learning

#### 3.1.1 Unified Architectures for Sequence Structure Semantics

Unified architectures for sequence-structure semantics aim to resolve the historical decoupling of protein language modeling and geometric reasoning by embedding both modalities within a single computational framework. Unlike traditional pipelines that rely on separate inverse-folding steps, these models utilize shared position encodings and joint tokenization schemes to guarantee precise residue-by-residue alignment between amino acid identities and their corresponding spatial coordinates. By treating continuous structural tokens and discrete sequence elements as interleaved inputs, the architecture learns a cohesive distribution where semantic function emerges directly from the interplay of primary sequence and tertiary fold.

Recent advancements employ transformer-based decoder-only or encoder-decoder configurations to model this joint distribution, often leveraging roto-translation invariant representations to ensure physical plausibility. These systems mitigate error accumulation inherent in local tokenization by constructing global representations where successive tokens contribute increasing levels of structural detail. Techniques such as masking strategies for unknown semantic identities and any-order autoregressive modeling allow the network to iteratively refine both sequence and structure simultaneously, effectively navigating the complex energy landscape without requiring external geometric constraints during the generation phase.

The integration of conditional guidance mechanisms further enhances these unified models, enabling the generation of novel proteins that adhere to specific functional motifs or enzymatic classes. By encoding motif residues as one-hot vectors and concatenating them with pairwise distance matrices, the architecture can perform motif-scaffolding and partial redesign while maintaining geometric consistency with the training distribution. This holistic approach not only improves the fidelity of generated candidates to natural protein families but also facilitates direct functional conditioning, allowing for the design of sequences that satisfy complex structural requirements and biological objectives in a single forward pass.

#### 3.1.2 Parameter Efficient Adaptation via Low Rank Updates

Parameter-Efficient Fine-Tuning (PEFT) via Low-Rank Adaptation has emerged as a critical strategy for customizing large protein language models without the prohibitive computational cost of full parameter retraining. By freezing the pretrained backbone weights and injecting trainable low-rank decomposition matrices into specific transformer layers, this approach drastically reduces the number of optimizable parameters. This methodology enables the rapid adaptation of foundational models, such as ESM-2, to downstream tasks like mutation effect prediction or conditional structure generation while preserving the rich evolutionary knowledge encoded during pretraining.

The technical implementation involves approximating weight updates $\Delta W$ as the product of two low-rank matrices, $BA$, where the rank $r$ is significantly smaller than the original model dimensions. During the fine-tuning process, only these adapter matrices are updated via gradient descent, while the original model parameters remain fixed. This architecture not only mitigates the risk of catastrophic forgetting but also facilitates efficient memory usage, allowing massive models to be deployed on consumer-grade hardware. Furthermore, it supports modular flexibility, where distinct adapter modules can be swapped to handle diverse objectives, such as switching from folding tasks to docking applications without maintaining separate full-scale model copies.

Recent investigations highlight that while standard fine-tuning of final layers offers some benefits, low-rank updates provide a more robust mechanism for aligning generative models with experimental fitness data. This is particularly relevant when integrating preference optimization techniques or conditioning on scarce experimental labels. By leveraging low-rank constraints, models can effectively navigate the fitness landscape to prioritize functional variants, achieving performance comparable to full fine-tuning with a fraction of the resource overhead. Consequently, this paradigm represents a scalable solution for iterative protein design cycles where rapid model specialization is essential.

#### 3.1.3 Context Aware Tokenization and Dynamic Vocabularies

Recent advancements in protein language models have shifted from static, residue-level tokenization toward context-aware schemes that encode structural and functional modalities directly into the vocabulary [8]. Traditional approaches treat amino acid sequences as discrete strings over a fixed alphabet, limiting the model's ability to capture long-range geometric dependencies. In contrast, modern frameworks employ joint tokenization strategies where structure is discretized via vector-quantized variational autoencoders into per-residue tokens representing local coordinate geometry. This unification allows a single transformer to process interleaved tracks of sequence, structure, and function, enabling generation conditioned on any subset of these modalities while maintaining consistency across the protein design space.

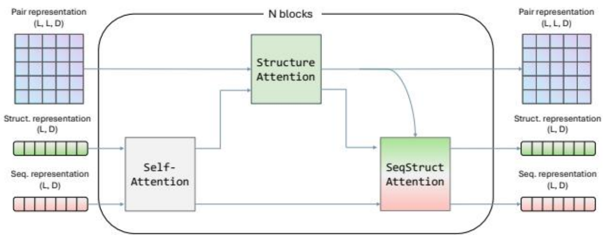


Figure 8: Chart from 'Elucidating the Design Space of Multimodal Protein Language Models' (arXiv:2504.11454)

To address the limitations of local neighborhood pooling, emerging methodologies introduce dynamic vocabularies capable of representing global structural contexts through coarse-to-fine hierarchies. Instead of mapping tokens to specific spatial regions, adaptive tokenizers utilize diffusion-based architectures to generate continuous structure tokens that are invariant to global rotations and translations. These dynamic representations allow the model to refine global embeddings iteratively, effectively capturing non-local interactions that are critical for folding stability. By replacing fixed embedding layers with learned modules that encode timestep information and augmented vocabulary sets, these systems facilitate the integration of noisy or partial structural data during both training and inference phases.

The flexibility of dynamic vocabularies further extends to handling variable sequence lengths and specialized functional annotations through mechanism like indel tokens and classifier-free guidance. Unlike standard masked language models that rely on a static set of substitutions, generative models equipped with augmented vocabularies can propose entirely new sequences by treating insertions, deletions, and functional labels as first-class tokens [9]. This approach supports complex tasks such as motif scaffolding and text-guided design, where the model must navigate a vast sequence space while adhering to strict geometric constraints. Consequently, context-aware tokenization transforms protein design from a local optimization problem into a holistic generative process capable of exploring diverse structural families.

## 3.2 Reinforcement Learning and Evolutionary Optimization

### 3.2.1 Policy Gradient Methods for Stability Maximization

Policy gradient methods have emerged as a pivotal strategy for maximizing protein stability by directly optimizing sequence generation policies against biophysical reward signals. Unlike supervised fine-tuning which relies on static datasets, these approaches formulate protein design as a Markov decision process where the agent iteratively constructs amino acid sequences to maximize an expected stability score. The core mechanism involves estimating the gradient of the expected reward with respect to the policy parameters, enabling the model to navigate the vast combinatorial sequence space towards regions of high thermodynamic fitness that may be underrepresented in training distributions.

To ensure training stability and prevent mode collapse, modern implementations frequently employ KL-divergence regularization against a frozen reference policy. This constraint limits the deviation from the pre-trained language model, preserving generalizable sequence features while steering generation toward stable variants. The objective function typically combines a token-level negative log-likelihood loss with a scaled advantage term derived from stability predictors, such as energy functions or deep learning-based scorers. By clipping negative advantages and retaining only top-performing trajectories within each batch, these methods effectively balance the exploitation of known stable motifs with the exploration of novel, high-fitness sequences.

Recent advancements have integrated these policy gradient frameworks with discrete diffusion models and transformer architectures to handle the non-differentiable nature of amino acid selection. Techniques such as the Gumbel-Softmax trick allow for the backpropagation of rewards through entire sampling paths, facilitating end-to-end optimization of global sequence properties. Furthermore, actor-critic architectures have been adapted to reduce variance in gradient estimates, ensuring robust convergence even when reward signals are sparse or noisy. This synergy between reinforcement learning and generative modeling represents a significant leap in designing proteins with enhanced thermostability and functional resilience.

### 3.2.2 Genetic Algorithms with Surrogate Fitness Landscapes

Genetic algorithms (GAs) have been adapted for protein design by replacing computationally prohibitive experimental fitness assays with surrogate models derived from deep learning architectures. In this paradigm, the fitness landscape is approximated using foundation models trained on vast sequence repositories or physics-based energy functions, enabling rapid evaluation of candidate variants. This substitution transforms the optimization process into a tractable search problem, where the surrogate acts as a differentiable or query-efficient proxy to guide evolutionary operators such as mutation, crossover, and selection toward regions of high predicted stability and function.

The integration of surrogate landscapes allows GAs to navigate the rugged, epistatic nature of protein fitness spaces more effectively than random mutagenesis or greedy walks. By leveraging metrics like designability and novelty, these algorithms can maintain population diversity while steering sequences away from local optima that satisfy biophysical constraints but lack functional utility. Advanced implementations often employ multi-objective optimization strategies to balance competing demands, such as solubility, immunogenicity, and binding affinity, generating Pareto-front solutions that span complex trade-offs inherent in biological engineering.

Despite their efficacy, reliance on surrogate models introduces challenges related to distributional shift and model bias, as generators may overfit to training data characteristics rather than true biological viability. To mitigate this, recent approaches incorporate active learning loops where high-performing in silico variants are experimentally validated and fed back to refine the surrogate's accuracy. Furthermore, techniques such as reinforcement learning and direct preference optimization are increasingly fused with genetic search mechanisms to align generated sequences with experimental feedback, ensuring that the explored sequence space remains both novel and biologically feasible.

#### 3.2.3 Reward Shaping for Multi Objective Protein Engineering

Reward shaping in multi-objective protein engineering addresses the inherent trade-offs between conflicting design criteria, such as stability, solubility, immunogenicity, and catalytic efficiency. Unlike single-objective optimization, which often converges to local maxima, multi-objective frameworks require reward functions that accurately reflect the complex, non-linear interdependencies of biological fitness landscapes. Traditional physics-based energy functions frequently fail to capture these global properties, necessitating the integration of learned reward models that can evaluate holistic protein characteristics rather than isolated structural features.

To overcome the sparsity of global rewards, advanced shaping techniques decompose high-level objectives into dense, intermediate signals that guide generative models during sequence or structure sampling. Methods leveraging reinforcement learning and direct preference optimization align generative policies with experimental fitness data, effectively bridging the gap between in silico predictions and wet-lab outcomes. By incorporating Pareto-based sampling strategies, these approaches generate diverse candidate sets that span the frontier of optimal trade-offs, allowing researchers to select designs that best balance competing constraints without sacrificing novelty or structural plausibility.

Furthermore, recent advancements utilize continuous surface representations and multimodal tokens to refine reward granularity, ensuring that partial states receive meaningful feedback even before full protein completion. This mitigates the reliability issues associated with calculating global metrics like binding affinity on incomplete structures. By dynamically adjusting reward weights based on evolutionary constraints and functional site importance, shaped reward mechanisms enable more efficient beam search and sampling procedures. Consequently, these techniques significantly enhance the success rates of de novo designs, facilitating the creation of functional proteins tailored for specific therapeutic and industrial applications.

### 3.3 Controllable Generation and Functional Steering

#### 3.3.1 Activation Steering for Property Specific Modulation

Activation steering offers a training-free mechanism to modulate protein generative models by directly intervening in latent representations during inference. Unlike reinforcement learning or fine-tuning, which require expensive parameter updates, this approach computes property-specific direction vectors within the activation space of pre-trained transformers. By adding a scaled steering vector to the hidden states at designated layers, the model's trajectory is biased toward desired biochemical attributes, such as enhanced binding affinity or thermal stability, while preserving the underlying structural validity learned during pre-training.

The efficacy of this modulation relies on accurately identifying orthogonal directions that correlate with specific functional properties without disrupting the model's generative coherence. Techniques often involve analyzing the gradient of a property predictor with respect to internal activations or utilizing contrastive sets of sequences to derive difference vectors. These vectors are then injected into the forward pass, effectively shifting the probability distribution over amino acid tokens. This process allows for continuous control via a guidance strength parameter, enabling a smooth trade-off between the magnitude of property enhancement and the likelihood of generating non-viable sequences.

Recent advancements have extended activation steering to handle multi-objective optimization by combining multiple steering vectors or employing iterative refinement strategies. This is particularly critical in protein design, where satisfying conflicting constraints, such as maximizing solubility while maintaining a specific fold, is challenging. By framing the steering problem as sampling from a Bayesian posterior where the pre-trained model acts as a prior and the steering vector approximates a likelihood term, researchers can achieve precise control over sequence characteristics. This method significantly reduces the computational overhead associated with retraining, facilitating rapid exploration of the sequence space for novel therapeutic candidates.

#### 3.3.2 Concept Bottlenecks for Interpretable Design Control

Concept bottlenecks serve as a critical architectural mechanism for enforcing interpretable control within generative design pipelines, particularly when addressing the inherent edit-based nature of biological engineering. Unlike black-box approaches that rely on latent space interpolation, concept bottlenecks explicitly constrain the generation process through human-understandable variables, such as specific structural motifs or functional sites. This alignment allows practitioners to iteratively introduce substitutions and indels to modify loops or binding interfaces while rigorously preserving the overall fold, thereby bridging the gap between abstract model outputs and synthetically accessible designs.

The implementation of these bottlenecks often involves integrating informative guidance mechanisms that select optimal starting points rather than relying on random noise initialization. By defining energy functions that prioritize backbone structure consistency, models can leverage causal attention layers to enhance design quality without necessitating excessive depth; empirical evidence suggests that a single causal layer often achieves optimal performance, whereas deeper configurations yield diminishing returns. This efficiency is crucial for maintaining high designability scores, as increasing sampling steps or model complexity does not consistently correlate with improved structural fidelity and may instead introduce hallucinations or reduce diversity.

Furthermore, concept bottlenecks facilitate a paradigm shift from pure prediction to controllable generation by enabling plug-and-play adaptation without retraining expensive foundation models. This approach supports uncertainty-calibrated oracles that distinguish confident predictions from artifacts, ensuring that generated variants remain within viable regions of the sequence space. By trading some representation fidelity for enhanced generative capacity and explicit control, these methods allow for the seamless integration of experimental feedback into design cycles, ultimately fostering more robust and operationally effective workflows for real-world enzyme and protein design campaigns.

#### 3.3.3 Preference Optimization Aligned with Energy Functions

Preference optimization aligned with energy functions represents a paradigm shift from traditional maximum likelihood estimation to reward-driven fine-tuning in protein design. By framing biophysical stability and functional fitness as explicit energy landscapes, these methods leverage Direct Preference Optimization (DPO) to align generative models with experimental data without requiring explicit reward modeling. The core mechanism involves constructing preference pairs where sequences exhibiting lower Rosetta energy scores or higher experimental fitness are designated as preferred over non-functional or high-energy variants, effectively steering the model distribution toward the global minimum of the physicochemical energy function.

Recent advancements integrate exact energy preference optimization with regularization terms to mitigate the collapse of sequence diversity often observed in affinity-driven sampling. Unlike standard reinforcement learning approaches that rely on unstable policy gradients, energy-aligned DPO utilizes a closed-form objective derived from the Bradley-Terry model, directly optimizing the policy to maximize the margin between preferred and dispreferred sequences based on their computed energy values. This approach allows for the precise incorporation of multi-objective constraints, such as balancing thermodynamic stability with solubility and immunogenicity, by defining composite energy functions that serve as the ground truth for preference ranking during the fine-tuning phase.

The efficacy of this alignment strategy is evident in its ability to navigate the rugged fitness landscapes characteristic of high-dimensional protein sequence spaces. By retaining a controlled degree of stochasticity through temperature-scaled sampling rather than relying on deterministic argmax selection, these models avoid local minima while converging toward viable conformations. Empirical results indicate that integrating biophysically informed energy functions into the preference loss yields variants with superior structural relaxation and functional performance compared to baseline generative models, establishing a robust framework for inverse folding that harmonizes deep generative capabilities with rigorous physical constraints.

### 3.4 Agentic Workflows and Benchmarking Ecosystems

#### 3.4.1 Tool Augmented Planning for Iterative Refinement

Tool-augmented planning transforms iterative protein design from static generation into a dynamic, closed-loop optimization process. By integrating external evaluators, such as property predictors or physics-based energy functions, directly into the sampling trajectory, generative models can navigate complex fitness landscapes more effectively than through unconditional sampling alone. This paradigm shifts the objective from merely satisfying distributional constraints to actively maximizing specific functional metrics, allowing the system to correct deviations and refine structural motifs in real-time without requiring expensive model retraining.

The core mechanism relies on treating the design workflow as a sequential decision-making problem where an agent proposes edits-ranging from single-point substitutions to indels-based on feedback from auxiliary tools. In each iteration, the current candidate sequence is assessed by these oracle models, and the resulting gradients or reward signals guide the subsequent unmasking or modification steps. This approach enables a conservative refinement strategy that prioritizes high-confidence substitutions while preserving the global fold, effectively balancing the tension between exploring novel sequence space and maintaining structural viability.

Recent advancements have extended this framework to include reinforcement learning objectives and preference-based alignment, further enhancing the model's ability to converge on optimal solutions. By dynamically adjusting the masking schedule and leveraging multi-step feedback loops, these systems demonstrate superior performance in tasks requiring precise interface optimization or motif scaffolding. The integration of such planning modules ensures that generative processes remain computationally tractable while achieving higher designability scores and functional accuracy compared to traditional one-shot generation methods.

#### 3.4.2 Retrieval Augmented Generation for Structural Priors

Retrieval Augmented Generation (RAG) for structural priors addresses the limitations of random noise initialization by leveraging informative guidance derived from existing biological data. Instead of sampling from a purely stochastic distribution, this approach defines an energy function across residues to select optimal starting points from a pool of noisy candidates. By retrieving secondary structure fragments identified via tools like DSSP from training datasets, the model ensures that initialized sequences and backbones maintain geometric similarities to natural proteins. This strategy effectively constrains the search space, providing the generative process with a biologically informed prior that enhances the quality and foldability of candidate structures.

The integration of retrieval mechanisms extends to multimodal foundation models capable of joint sequence-structure co-generation. In these frameworks, structural tokens are sampled first, often utilizing chain-of-thought inference to establish a backbone before sequence completion. By incorporating motif-level structural retrieval into sequence representations, models can condition generation on specific functional tags or partial structures. This decoupling of conditioning from the sampling process allows for the preservation of functional mechanisms while exploring novel sequence spaces. Consequently, the model learns underlying sequence-structure mapping principles implicitly, facilitating tasks such as inverse folding and de novo design without explicit supervision for each sub-task.

Evaluation of these retrieval-augmented approaches demonstrates superior performance in maintaining distributional similarity and conditional consistency compared to standard one-stage generative models. Metrics such as Maximum Mean Discrepancy and Mean Reciprocal Rank confirm that generated sequences adhere to natural background frequencies and global statistics inherent in pre-trained weights. Furthermore, by ranking generated structures based on RMSD values against target motifs, the method identifies candidates with high structural fidelity. As underlying backbone generative models improve, the efficacy of guided generation is expected to increase concomitantly, offering a robust pathway for designing proteins with specific functional properties and enhanced stability.

#### 3.4.3 Holistic Evaluation Metrics for Novelty and Diversity

Holistic evaluation of generative protein models necessitates a unified framework that simultaneously assesses novelty and diversity alongside designability, rather than treating them as isolated objectives. Novelty is rigorously quantified by measuring structural deviation from known biological repositories, such as the PDB or AFDB-SwissProt, using alignment tools like Foldseek or TM-align. A generated structure is typically deemed novel if its maximum TM-score against all reference entries falls below a stringent threshold, often 0.5, ensuring the model explores uncharted regions of the structural universe rather than merely memorizing existing folds.

Diversity metrics complement novelty by evaluating the variability within the set of generated candidates themselves, preventing mode collapse where a model produces identical high-quality structures repeatedly. This is commonly operationalized by computing pairwise inter-chain TM-scores or sequence dissimilarity among the top-k designable samples, where lower average similarity scores indicate a broader exploration of the fitness landscape. Advanced approaches further refine this by employing hierarchical clustering on designable outputs to count distinct structural clusters, thereby capturing the richness of the generated ensemble beyond simple pairwise averages.

Crucially, these metrics must be interpreted holistically, as optimizing for diversity or novelty in isolation can compromise structural validity and functional fitness. Effective evaluation strategies often employ composite scores or Pareto frontiers to balance the trade-off between generating unique, varied structures and maintaining high designability, indicated by metrics like pLDDT and PAE. By benchmarking against natural protein baselines and utilizing multi-objective optimization techniques, researchers can ensure that generative models do not sacrifice biological plausibility for the sake of statistical novelty, ultimately validating their utility in discovering functional, non-natural biomolecules.

# 4 Deep Generative Architectures and Hybrid Frameworks for Biomolecular Design

## 4.1 Graph Neural Networks and Geometric Deep Learning

### 4.1.1 Equivariant Message Passing for Interaction Modeling

Equivariant message passing mechanisms have emerged as a foundational component for modeling biomolecular interactions, ensuring that predicted physical properties remain consistent under Euclidean transformations. By explicitly separating node features from spatial coordinates, these architectures utilize relative position vectors within the message aggregation function to preserve E(3) or SE(3) equivariance. This design guarantees that rotating or translating the input structure results in a correspondingly transformed output, a critical requirement for accurately capturing the geometry-dependent nature of residue-residue and protein-ligand binding interfaces without relying on data augmentation.

In the context of interaction modeling, equivariant graphs facilitate the iterative refinement of both semantic embeddings and 3D coordinates through coupled update rules. Messages are constructed using invariant scalar distances and equivariant vector differences, allowing the network to reason about steric constraints, orientation preferences, and long-range electrostatic effects simultaneously. Advanced implementations often incorporate attention mechanisms that weigh neighbor contributions based on geometric compatibility, enabling the model to distinguish between specific binding hotspots and non-interacting surface regions while maintaining strict physical symmetries throughout the deep hierarchy of layers.

Recent developments extend this paradigm to support complex multi-chain systems and dynamic conformational changes by integrating hierarchical residue interaction modeling with equivariant constraints. These frameworks achieve iterative sequence-structure co-optimization, where latent representations guide the generation of plausible binding geometries while adhering to rigid body transformations. By operationalizing reasoning as explicit geometric commitments rather than abstract latent mappings, equivariant message passing significantly reduces the search space for valid interaction modes, thereby enhancing the reliability of de novo design and docking predictions in high-dimensional structural landscapes.

#### 4.1.2 Topological Data Analysis for Fitness Landscape Mapping

Topological Data Analysis (TDA) offers a rigorous mathematical framework for characterizing the high-dimensional geometry of protein fitness landscapes, transcending the limitations of static structural datasets. By leveraging persistent homology, researchers can quantify the connectivity and voids within sequence space, effectively mapping the complex, nonlinear relationships between amino acid variations and functional properties. This approach identifies robust topological features, such as connected components representing distinct functional basins and loops indicating evolutionary pathways, which are often obscured by noise in traditional energy function approximations.

The integration of TDA with machine learning facilitates the navigation of vast combinatorial spaces by distinguishing global topological accuracy from local geometric distortions. Metrics such as the predicted Template Modeling score serve as proxies for global fold correctness, enabling the differentiation between sequences trapped in local optima and those capable of accessing novel functional combinations. Through the analysis of normal mode shapes and conformational dynamics, TDA-informed models can smooth oscillating fitness signals, shifting probability distributions toward regions of high accuracy and stability while preserving the essential diversity required for de novo design.

Furthermore, this methodology supports the construction of reinforcement learning ecosystems that jointly optimize sequence, structure, and function. By decoding evolutionary covariation signals and geometric constraints, TDA guides generative processes to avoid biases toward specific secondary structures, such as alpha-helices, ensuring the robust generation of complex beta-sheet topologies. This synergy between topological abstraction and data-driven prediction establishes a "predict-design-validate" paradigm, allowing for the efficient exploration of epistatic interactions and the discovery of proteins with unprecedented structural fidelity and functional novelty.

#### 4.1.3 Global Context Awareness in Protein Graph Representations

Global context awareness in protein graph representations addresses the limitation of local message-passing mechanisms, which often fail to capture long-range dependencies critical for defining global topology and fold stability. While standard Graph Neural Networks (GNNs) excel at modeling immediate spatial neighborhoods through iterative aggregation, they inherently possess a spatial locality bias that obscures distant interactions between secondary structure elements. To mitigate this, recent architectures integrate global attention mechanisms or hierarchical pooling strategies, enabling nodes to access information from the entire structural manifold rather than relying solely on truncated Euclidean neighborhoods.

The incorporation of global context is particularly vital for distinguishing functional motifs that exhibit diverse geometric realizations across different topological scaffolds. Without explicit global conditioning, generative models treating structural tokenization via discrete diffusion often encounter mode collapse, interpreting distinct functional instantiations as competing probabilities rather than context-dependent variations. Advanced frameworks resolve this ambiguity by decoupling molecular understanding from geometric generation, utilizing multimodal experts to inject high-level semantic constraints into the latent space. This ensures that local structural motifs are generated in coherence with the overarching fold, preserving both functional integrity and thermodynamic stability.

Furthermore, achieving SE(3) invariance while maintaining global sensitivity requires sophisticated encoding of pairwise geometric features alongside scalar residue attributes. Modern approaches employ graph transformers that merge the inductive bias of GNNs with the expressive power of self-attention, allowing for the direct modeling of all-pair interactions regardless of spatial distance. By encoding inter-residue distances and orientations within a global attention matrix, these models effectively capture the complex interplay between local chemistry and global architecture. This synthesis facilitates accurate sequence-structure co-optimization, ensuring that designed proteins satisfy both local steric constraints and global topological requirements essential for novel function.

## 4.2 Variational Autoencoders and Latent Space Modeling

### 4.2.1 Continuous Latent Manifolds for Sequence Interpolation

Continuous latent manifolds provide a structured geometric framework for modeling protein sequence distributions, enabling smooth transitions between discrete amino acid strings. By mapping high-dimensional sequence data into a compressed, real-valued latent space via variational autoencoders or denoising autoencoders, these models capture intrinsic functional dependencies and evolutionary constraints. This transformation allows the representation of complex sequence families as continuous topological surfaces, where proximity in the latent domain correlates with structural and functional similarity, effectively bridging the gap between discrete token spaces and differentiable optimization landscapes.

Sequence interpolation within these manifolds facilitates the generation of novel, viable protein variants by traversing geodesic paths between known functional endpoints. Unlike discrete mutation strategies that often encounter rugged fitness landscapes, continuous interpolation leverages the smoothness of the latent prior to synthesize intermediate sequences that maintain global fold stability, as estimated by metrics such as the predicted Template Modeling score. The generative process typically involves sampling points along the vector connecting two latent embeddings, which are then decoded back into sequence space, yielding non-linear combinations of parental traits that preserve critical long-range interactions and torsional angle restrictions.

Advanced implementations integrate self-conditioning mechanisms and equivariant diffusion processes to refine these interpolated trajectories, ensuring that generated sequences adhere to physical constraints while exploring regions beyond the training distribution. By incorporating rotary positional embeddings and attention-based architectures, modern frameworks generalize effectively to sequences of varying lengths, mitigating the locality bias inherent in recurrent approaches. Consequently, navigating these continuous manifolds not only accelerates the discovery of optimized sequences through gradient-based methods but also provides a scalable alternative to multiple sequence alignments for predicting the effects of mutations on protein function and stability.

### 4.2.2 Vector Quantization for Discrete Structural Tokenization

Vector quantization serves as a critical mechanism for discretizing continuous protein backbone coordinates into fixed-size vocabulary tokens, enabling the application of discrete diffusion models to structural generation. By mapping local structural motifs of amino acids to discrete indices within a predefined codebook, this approach transforms complex geometric data into a sequence of categorical variables suitable for language modeling paradigms. This tokenization process effectively captures local structural contexts while reducing the infinite continuous conformational space into a manageable discrete domain, facilitating the joint modeling of sequence and structure distributions.

However, optimizing these models for functional constraints presents significant challenges due to training ambiguities inherent in the quantization process. The discrepancy between continuous latent representations and their discrete counterparts often leads to optimization instability, exacerbated by the bias in straight-through estimators and the risk of codebook collapse where only a subset of embeddings is utilized. Traditional methods relying on explicit codebook learning struggle with computational costs associated with pairwise distance calculations, particularly for long sequences, necessitating advanced regularization techniques to ensure full codebook capacity utilization and stable gradient propagation during training.

To mitigate these issues, recent frameworks integrate supervision directly into the vector-quantized structure tokenizer rather than relying solely on discrete output supervision. By employing look-up free quantization or similar variants, models can bypass explicit codebook maintenance while maintaining high structural fidelity. This allows for iterative sequence-structure co-optimization where discrete structural tokens condition sequence generation, effectively bridging the gap between vibrational dynamics prediction and de novo design. Such architectures enable the model to learn robust mappings from natural language descriptions or normal mode vectors to discrete structural backbones, enhancing the generation of proteins with complex topological features.

#### 4.2.3 Bayesian Flow Networks for Profile Guided Design

Bayesian Flow Networks (BFNs) have emerged as a potent framework for profile-guided protein design by operating directly on discrete state-spaces, thereby overcoming the limitations of continuous relaxations inherent in traditional diffusion models. Unlike standard flow-based approaches that require intricate dequantization techniques, BFNs utilize a continuous-time Markov process to define probability flows over amino acid sequences, enabling exact likelihood evaluation and efficient sampling. This architectural choice is particularly advantageous for co-design tasks, where the model must simultaneously optimize sequence identity and structural conformation while adhering to complex evolutionary profiles derived from multiple sequence alignments.

In the context of profile-guided design, BFNs facilitate a rigorous integration of prior biological knowledge through conditional guidance mechanisms. By treating evolutionary profiles as conditioning inputs, the flow network learns to navigate the high-dimensional fitness landscape, steering the generative process toward regions that satisfy both thermodynamic stability and functional constraints. The invertible nature of the transformation allows for bidirectional mapping between latent noise distributions and valid protein sequences, ensuring that generated candidates maintain high designability scores. Furthermore, the probabilistic formulation supports Bayesian optimization strategies, allowing researchers to quantify uncertainty and iteratively refine designs based on sparse experimental feedback or specific dynamical signatures.

Recent advancements have extended this paradigm to joint sequence-structure co-design, where BFNs are coupled with geometric modules to enforce SE(3) equivariance during backbone generation. This hybrid approach enables the model to capture long-range dependencies and multimodal distributions essential for designing novel folds that deviate from known structural templates. By leveraging gradient-based optimization through the flow objective, these networks can effectively balance competing objectives, such as binding affinity and solubility, within a unified training regime. Consequently, Bayesian Flow Networks represent a significant step toward rational, data-efficient protein engineering, offering a robust mathematical foundation for generating diverse, functional biomolecules tailored to precise specification profiles.

### 4.3 Generative Adversarial Networks and Energy Based Models

#### 4.3.1 Adversarial Training for Functional Sequence Synthesis

Adversarial training has emerged as a pivotal paradigm for functional sequence synthesis, leveraging the competitive dynamics between generators and discriminators to navigate vast protein sequence spaces [10]. Unlike energy-based models reliant on heuristic searches, Generative Adversarial Networks (GANs) learn implicit distributions of natural sequences to produce novel candidates with high structural fidelity. Early implementations, such as ProteinGAN, demonstrated that adversarial frameworks could generate functional malate dehydrogenases exhibiting near wild-type activity, effectively expanding the accessible functional sequence space beyond natural evolutionary constraints while maintaining biophysical plausibility.

$$\mathbf{x} \xrightarrow{\text{encoder}} \mathbf{e} \xrightarrow{\text{quantizer}} \mathbf{z} \xrightarrow{\text{decoder}} \tilde{\mathbf{x}},$$

Figure 9: Chart from 'DPLM-2: A Multimodal Diffusion Protein Language Model' (arXiv:2410.13782)

The methodology typically employs a generator to map latent vectors to amino acid sequences and a discriminator to distinguish synthetic outputs from natural homologs, often conditioned on specific functional labels or structural motifs. This likelihood-free approach allows for direct optimization of sequence properties without explicit probability density estimation. Recent advancements integrate protein language models as backbones within the adversarial framework, enabling the generator to inherit robust priors regarding sequence stability and foldability. By fine-tuning projection layers that align structural features with language representations, these hybrid architectures achieve superior controllability, ensuring generated sequences satisfy both global functional requirements and local geometric constraints.

Despite their promise, adversarial approaches face challenges related to mode collapse and the difficulty of capturing diverse sequence realizations underlying a single function. One-step generation strategies sometimes yield degraded functionality due to the complexity of learning diverse functional manifolds, whereas two-step paradigms attempting to ground function in explicit structure may compromise foldability. To mitigate these issues, contemporary research focuses on stabilizing training through Wasserstein distances and incorporating auxiliary predictors that evaluate dynamic accuracy or vibrational modes during the adversarial loop. These refinements aim to balance the trade-off between sequence diversity and functional specificity, pushing the boundaries of de novo protein design.

#### 4.3.2 Boltzmann Generators for Thermodynamic Ensemble Sampling

Boltzmann generators represent a paradigm shift in sampling thermodynamic ensembles by leveraging normalizing flows to learn bijective mappings between complex many-body configurations and tractable latent distributions. Unlike traditional Markov Chain Monte Carlo methods that suffer from slow mixing across high energy barriers, these deep learning architectures directly approximate the Boltzmann distribution through reweighting techniques. By training on limited molecular dynamics trajectories, the model learns to generate uncorrelated equilibrium states efficiently, effectively bypassing the kinetic traps that hinder conventional sampling in rugged protein energy landscapes.

The core mechanism involves optimizing a neural network to minimize the Kullback-Leibler divergence between the generated distribution and the target thermodynamic ensemble defined by physical force fields. This approach enables the direct generation of diverse protein conformations that satisfy detailed balance without requiring iterative relaxation steps [11]. The integration of SE(3) equivariance ensures that the generated structural ensembles respect fundamental physical symmetries, allowing for the accurate modeling of intrinsically disordered regions and large-scale conformational transitions that are critical for understanding protein function and allostery.

Recent advancements have extended this framework to discrete state spaces and conditional generation tasks, facilitating the co-design of amino acid sequences and backbone structures. By incorporating energy minimization constraints directly into the flow architecture, Boltzmann generators can produce globally feasible sequences that avoid local minima associated with high-frequency amino acid patterns. This capability bridges the gap between static structure prediction and dynamic stability assessment, offering a robust platform for designing functional proteins with tailored thermodynamic properties and enhanced conformational diversity.

$$E(\mathbf{x}) = \frac{\alpha}{2} \sum_{(i,j)\in\mathcal{E}} ||\mathbf{x}_i - \mathbf{x}_j||^2$$

Figure 10: Chart from 'EigenFold: Generative Protein Structure Prediction with Diffusion Models' (arXiv:2304.02198)

#### 4.3.3 Energy Based Priors for Inverse Folding Optimization

Energy-based priors have emerged as a critical mechanism for refining inverse folding optimization by explicitly encoding physical constraints into the sequence generation process [6]. Unlike traditional methods that rely solely on statistical potentials derived from natural sequences, energy-based models (EBMs) define a learnable scalar energy landscape over the joint space of sequences and structures. This approach allows the model to penalize configurations that violate thermodynamic stability or steric feasibility, effectively guiding the sampling trajectory toward low-energy states that correspond to foldable, functional proteins. By integrating these priors, generative frameworks can overcome the limitations of purely data-driven approaches, which often struggle to generalize to non-canonical folds or novel functional motifs absent in training distributions.

The integration of EBMs typically operates by augmenting standard inverse folding architectures, such as graph neural networks or transformers, with an additional energy term that reflects physical realism. During optimization, the sequence is iteratively updated to minimize a composite objective function comprising both the negative log-likelihood of the sequence given the backbone and the predicted energy score. This dual-objective strategy mitigates the risk of converging to "safe" local minima characterized by repetitive motifs or excessive electrostatic patterns, which frequently plague conventional inverse folding tools. Furthermore, by leveraging techniques like Langevin dynamics or contrastive divergence, these models can efficiently explore the rugged energy landscape, ensuring that the generated sequences possess not only high recovery rates but also robust folding properties consistent with quantum mechanical arguments and molecular dynamics simulations.

Recent advancements have extended this paradigm to incorporate dynamic conformational sampling, where energy priors are conditioned on specific normal modes or vibrational signatures rather than static backbones alone. This evolution enables the design of proteins with tailored dynamic behaviors, addressing the longstanding challenge of mapping sequence to function through intermediate structural dynamics. By treating the energy function as a differentiable prior within end-to-end diffusion or flow-matching pipelines, researchers can achieve a seamless synergy between geometric plausibility and functional specificity. Consequently, energy-based priors serve as a foundational bridge between physics-based simulation and deep generative modeling, significantly enhancing the reliability of de novo protein design for complex applications such as enzyme engineering and binder development.

## 4.4 Multimodal Integration and Hybrid Architectures

### 4.4.1 Cross Modal Alignment of Text Structure and Sequence

Recent advancements in protein design have shifted toward unified generative frameworks that jointly model sequence, structure, and function through cross-modal alignment. Unlike traditional approaches that treat these modalities sequentially, modern architectures employ interleaved decoding strategies to enhance functional modeling via structural integration. By incorporating explicit sequence constraints during the generation process, these models ensure foldability while exploring the geometric space with sufficient flexibility [6]. This paradigm allows for the simultaneous optimization of amino acid identities and spatial coordinates, effectively bridging the gap between linear textual representations and three-dimensional structural realities.

To achieve precise alignment, sophisticated mechanisms such as cross-attention fusion and distributional alignment via Latent Space Functional Supervision (LSFS) have been introduced. Cross-attention enables the precise integration of contextual sequence information with structural features, consistently outperforming alternative fusion methods in accuracy and interaction quality. Furthermore, LSFS significantly improves the alignment of generative distributions with natural functional protein spaces, as evidenced by substantial gains in Mean Reciprocal Rank and reductions in Maximum Mean Discrepancy. These techniques ensure that the generated sequences not only adhere to physical folding laws but also capture the inherent functional semantics required for biological activity.

The efficacy of these alignment strategies is further bolstered by the incorporation of evolutionary information and global decoding modules. While traditional methods rely heavily on computationally expensive Multiple Sequence Alignments, newer approaches utilize dense sequence retrievers or position-specific scoring matrices to inject evolutionary constraints without prohibitive costs. Additionally, introducing global Markov Random Field-based decoding modules enhances sequence-structure consistency within atom-level representation schemas. By iteratively refining residue types and updating coordinates, these systems achieve superior self-consistency, enabling the design of diverse protein libraries that maintain high structural stability and functional coherence.

### 4.4.2 Divide and Assemble Strategies for Complex Topologies

Divide and assemble strategies address the computational intractability of generating complex protein topologies by decomposing large-scale architectural problems into manageable sub-tasks. This paradigm typically involves partitioning multi-chain complexes or symmetric architectures into independent structural modules or repeating units, which are generated or optimized individually before being reassembled into a holistic structure. By leveraging stochastic decoding and coupled co-optimization of intra-chain and inter-chain equivalent residues, these methods significantly enhance geometric precision and design efficiency for repetitive structures. The approach effectively mitigates the curse of dimensionality inherent in simultaneous whole-complex generation, allowing for the exploration of vast conformational spaces while maintaining strict topological constraints.

The reassembly phase relies on sophisticated dynamic locking mechanisms and interface-aware refinement to ensure physical feasibility and stability across module boundaries. Advanced frameworks integrate diffusion models with inverse folding algorithms to transition seamlessly from topological scaffold generation to functional sequence optimization, bridging the gap between geometric primitives and chemical reality. This synergy enables the precise localization of interfaces and the optimization of interaction hotspots, ensuring that the assembled complex retains the desired binding modes and energetic profiles. Consequently, these strategies facilitate the design of higher-order symmetric architectures and de novo binders that were previously inaccessible to rigid, monolithic design pipelines.

Despite these advances, current divide and assemble approaches often treat backbone geometry as static, lacking explicit conditioning on protein dynamics and conformational flexibility. While metrics such as the predicted Template Modeling score provide reliable estimates of global fold accuracy for assembled structures, they do not fully capture the dynamic specificity required for functional efficacy in obligate complexes. Future iterations of these strategies must incorporate normal mode analysis and molecular dynamics simulations directly into the generative process to account for the elasticity and hierarchical movements essential for biological function. Integrating dynamic constraints will be critical for advancing from static structural prediction to the design of responsive, functional protein machines.

#### 4.4.3 Simulation Free Training via Ensemble Emulators

Simulation-free training via ensemble emulators addresses the prohibitive computational cost of molecular dynamics by substituting explicit physics simulations with learned surrogate models. These emulators approximate the statistical mechanical ensembles of protein conformations, enabling the efficient sampling of equilibrium states without solving Newton's equations of motion iteratively. By training on high-resolution data derived from limited simulations or experimental structures, these models learn to map sequence inputs directly to probability distributions over structural states, effectively capturing the underlying energy landscape.

The architecture of such emulators often leverages generative frameworks, including normalizing flows and diffusion models, to construct a differentiable approximation of the Boltzmann distribution. Instead of relying on force fields to guide conformational search, the emulator predicts an ensemble of plausible structures for a given sequence, allowing for the rapid evaluation of thermodynamic stability and foldability. This approach facilitates the integration of physical constraints, such as conservation laws and symmetry, directly into the loss function, ensuring that generated samples adhere to biophysical principles while maintaining computational tractability.

Deploying these emulators within a closed-loop design framework significantly accelerates protein engineering workflows. By acting as a high-throughput filter, the ensemble emulator evaluates vast candidate libraries generated by sequence models, identifying designs with favorable energetic profiles before any wet-lab validation. This strategy not only mitigates the risk of selecting non-foldable sequences but also enriches the training data for subsequent iterations with high-confidence synthetic examples. Consequently, simulation-free training bridges the gap between data-driven generative capabilities and rigorous physical validity, offering a scalable path toward functional protein design.

## 5 Future Directions

Despite the remarkable progress in diffusion-based protein design, several critical limitations persist that hinder the widespread adoption of these technologies for complex therapeutic and industrial applications. Current models often struggle with the seamless integration of discrete sequence optimization and continuous structural generation, frequently treating them as decoupled stages rather than a unified probabilistic process. This separation can lead to generated backbones that are geometrically sound but lack corresponding low-energy amino acid sequences, resulting in poor designability and experimental failure rates. Furthermore, while geometric equivariance ensures physical consistency, existing frameworks frequently lack explicit modeling of long-range electrostatic interactions and solvent effects, which are crucial for determining functional specificity and stability in physiological environments. The computational cost associated with sampling high-fidelity structures, particularly when incorporating rigorous physical guidance or handling large multi-chain complexes, remains prohibitive for high-throughput screening, limiting the exploration of the vast non-native sequence space.

Future research must prioritize the development of fully joint sequence-structure diffusion models that operate on hybrid discrete-continuous manifolds, enabling the simultaneous co-evolution of backbone geometry and side-chain identity. This entails advancing mathematical formulations that can handle discrete amino acid types within the same equivariant framework used for continuous coordinates, potentially leveraging masked diffusion strategies that dynamically update both modalities during the denoising trajectory. Additionally, there is a pressing need to integrate multi-scale physical simulations directly into the generative loop, moving beyond static energy functions to include dynamic solvation models and explicit electrostatic potentials. Exploring hierarchical generation schemes could also address the challenge of designing large assemblies and symmetric oligomers by decomposing the problem into local motif generation and global arrangement optimization. Finally, improving sample efficiency through advanced flow matching techniques and distillation methods will be essential to reduce the computational burden, allowing for the rapid generation of diverse candidate libraries suitable for experimental validation.

The realization of these future directions promises to fundamentally transform the landscape of synthetic biology and drug discovery. By achieving robust joint sequence-structure design, researchers will be able to engineer novel enzymes with catalytic efficiencies surpassing natural evolution and create therapeutic proteins with enhanced stability and reduced immunogenicity. The ability to accurately model complex physical interactions will facilitate the design of precise molecular machines and targeted binders for previously undruggable disease targets, accelerating the timeline from computational concept to clinical application. Ultimately, overcoming current limitations will shift protein engineering from a trial-and-error endeavor to a predictive science, unlocking the potential to fabricate bespoke biomaterials and life-saving therapeutics tailored to specific human needs with unprecedented precision and speed.

## 6 Conclusion

This survey has provided a comprehensive examination of the transformative role of diffusion models and protein language models in the realm of de novo protein design. We dissected the mathematical foundations underpinning these architectures, highlighting the critical shift from Euclidean assumptions to geometrically rigorous frameworks that operate on SE(3) Lie groups, toroidal manifolds, and rotational spaces. By enforcing equivariance and respecting the intrinsic curvature of protein geometry, modern generative models achieve superior sample quality and physical validity compared to earlier variational or adversarial approaches. Furthermore, our analysis of conditional generation strategies revealed how force-guided sampling and constraint anchoring effectively bridge the gap between statistical learning and biophysical reality. These methods enable the precise steering of generative trajectories toward low-energy conformations and allow for the rigid scaffolding of functional motifs, thereby addressing the core challenges of stability and functionality. Finally, the integration of discrete sequence recovery with continuous backbone generation marks a significant advancement toward holistic joint design, offering a unified pathway to navigate the vast landscape of synthetic biomolecules.

The significance of this review lies in its systematic synthesis of a rapidly fragmenting field, offering a structured taxonomy that clarifies the distinct architectural innovations required for handling the unique geometric and discrete nature of proteins. Unlike broader surveys on deep learning in biology, this work provides a deep technical dive into the specific mechanisms of geometric deep learning and probabilistic modeling that define the current state-of-the-art. By critically evaluating the synergy between equivariant neural networks, flow matching techniques, and energy-based guidance, we establish a clear benchmark for assessing model performance and theoretical soundness. This consolidation serves as an essential resource for researchers, distilling complex mathematical concepts into actionable insights and identifying the precise algorithmic components that drive success in generating experimentally viable protein structures. It effectively maps the transition from analyzing natural evolution to inventing synthetic biology, underscoring the potential of these tools to accelerate discoveries in therapeutics and materials science.

Looking forward, the field stands at a pivotal juncture where algorithmic sophistication must be matched by rigorous experimental validation and scalability. While current models demonstrate remarkable prowess in generating novel folds, future research must address the remaining challenges in modeling large-scale multi-chain complexes, dynamic conformational ensembles, and the intricate coupling of sequence and function beyond static structures. There is an urgent need for standardized benchmarks that evaluate not only structural novelty but also expressibility, solubility, and specific functional metrics in wet-lab settings. We call upon the community to foster tighter feedback loops between computational prediction and experimental characterization, ensuring that generative models are iteratively refined by real-world biological data. As these technologies mature, they promise to unlock a new era of protein engineering, moving from the design of individual domains to the creation of entire synthetic biological systems. The convergence of geometric reasoning, physical priors, and massive-scale data processing will ultimately define the next generation of intelligent tools capable of solving some of biology's most profound design problems.

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